Diagnostic check: six questions, 10 minutes

Answer without consulting the course. Indicate the questions for which assistance is required. diagnostic check is not graded.

1. Derivative from x^2 in three?

2. Domain of $\dfrac{1}{x-2}$?

3. Sign of ($x-1)(x+2)$?

4. Develop ($x^2+1)(x-2)$.

5. For $f(x)=x^2$, compare $f(-x)$ and $f(x)$.

6. If $f(1)=2,\ f\prime(1) = 0$, tangent equation?

After the diagnostic check. Retrieve the concepts indicated with the teacher; repeat a question and then explain the rule used.

Speaking and reasoning. Present a table of variations by distinguishing between the sign of f and its derivative. Explain why the bounds of an optimization problem need to be examined.

Search before formalizing

Can a increasing function have a horizontal tangent? Look for an example; can it Have a strictly negative derivative at an inner point?

Organization. Individual research 5 min, confrontation 2 5 min, pooling 5 min. Keep a trace of a test, even if not completed. A conjecture must then be justified.